

Twin Comp

Two classic compressors in one rack - a VCA and an Expessor - in either order.



Overview

Twin Comp puts two very different compressors side by side. Unit A is a VCA compressor in the spirit of the classic late-70s RMS boxes: it measures loudness rather than peaks, has a soft knee, and its timing follows the music. Its COMPRESSION knob runs from 1:1 through INF:1 and on into negative ratios.

Unit B is an Expessor in the spirit of the classic 80s expander-compressors: peak-sensing with a smooth knee, with DRIVE, RATIO, ATTACK and RELEASE, an AUTO timing mode, HF EXP to keep the top open under heavy compression, SPR to firm up the low end and LF CUT to stop the bass from pumping it.

ORDER sets which unit comes first, and the two units swap places on the panel to match. MIX blends in the dry signal for parallel compression.

Reach for it for

- Drum buses that want punch and control (A into B)
- Vocals that need firm, open-sounding compression (B with HF EXP)
- Bass that needs a tight, steady bottom (SPR)
- Gentle glue on a mix (both units lightly)
- Pumping and reverse-dynamics effects (A past INF)

Controls

Drag a knob up or down to turn it (hold Shift for fine moves), or double-click to reset it. The value shows while the mouse is over a knob, and the display shows the control you are changing.

The top strip

Display	Shows the preset number and name, or for a moment the control you have just changed. Click it for the preset list.
< and >	Step to the previous or next preset.
ORDER	Click to swap the order: A > B or B > A. The two units change places on the panel so the first one is on the left. The key's light shows while it is held.
MIX	Parallel compression, 0 - 100%: 100% is fully compressed, lower settings blend the dry signal back in.
DRIFT	An uneven supply under both VCAs, 0 - 100% (20% by default): each unit's gain wobbles very slightly (up to +/-0.15 dB at 100%) and unit A's colour moves with it. It stays level-neutral.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper that holds the peaks at -0.3 dBFS (no latency), or a lookahead limiter that holds them at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

A - VCA COMPRESSOR (true RMS)

ON	Switches unit A in or out.
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BELOW / KNEE / ABOVE	Three lights showing where the level sits against the threshold: green below it, amber in the soft knee, red above it.
THRESHOLD	The level where compression starts, -40 to 0 dB. Because the unit measures loudness, it relates to the average level rather than the peaks.
COMPRESSION	The ratio. From fully left (1:1) it rises to INF:1 at 80% of its travel; past that it becomes negative, down to -2:1 at the end: the louder the input, the quieter the output. The value is shown as a ratio.
OUTPUT	Unit A's make-up gain, -20 to +20 dB.
SOFT KNEE	On (default): the ratio eases in over 10 dB around the threshold. Off: a hard knee, firmer and more obvious.
GR	The column of lights shows unit A's gain reduction, 0 to 20 dB.

There are no attack or release controls on unit A: the further the signal goes over the threshold, the faster it reacts (about 15 ms for 10 dB over, 5 ms for 20 dB), and it recovers at a steady 120 dB per second. It adds a whisper of 2nd harmonic that grows with the gain reduction.

B - EXPRESSOR (peak)

ON	Switches unit B in or out.
DRIVE	How hard the signal is pushed into the compressor, 0 to +40 dB. More drive means more compression (it sets the threshold from 0 down to -40 dB).
RATIO	1:1 to 20:1.
ATTACK	0.1 - 50 ms. Reads AUTO when AUTO is on.
RELEASE	30 ms - 3 s. Reads AUTO when AUTO is on.
OUTPUT	Unit B's make-up gain, -20 to +20 dB.

AUTO	Automatic, smooth optical-style timing: a quick attack and a two-stage release that follows the programme. ATTACK and RELEASE are not used.
SPR	Rotates the phase of the low end against the top (two all-pass stages at 180 Hz) for a tighter, firmer bottom.
LF CUT	A 150 Hz high-pass on what the compressor listens to only, so bass peaks no longer pump it. The bass itself is not filtered.
HF EXP	0 - 100%: lifts the top end (above about 3.5 kHz) in proportion to the gain reduction, up to +6 dB, so heavy compression stays open and bright.
GR	The column of lights shows unit B's gain reduction, 0 to 20 dB.
OUT	The output level of the whole plug-in, -30 dBFS to 0.

A light between the two units glows when both are on.

Presets

A preset sets the order, both units, which units are on, and MIX. DRIFT and the OFF / CLIP / LIMIT keys are left as they are.

Init	Both units on, A > B, at their starting values.
Drum Bus	Unit A only at 4:1 with a soft knee and 4 dB of make-up.
Vocal Expressor	Unit B only, 4:1 with AUTO timing, LF CUT and HF EXP, 5 dB of make-up.
Bass Tight	A > B: A with a hard knee, then B with SPR for a firm bottom.
Mix Glue	B > A: gentle 2:1 Expressor with AUTO, SPR and LF CUT, then light VCA compression.

Parallel Smash	A > B, both working hard (A with a hard knee, B at 8:1 with fast timing), blended in at 40%.
Acoustic Natural	Unit B only, 2.5:1 with AUTO timing, LF CUT and a little HF EXP.
Negative Pump	Unit A only, past INF for pumping and reverse-dynamics effects.

Tips

- A > B lets the VCA level the performance and the Expressor catch the peaks; B > A lets the Expressor shape the tone first and the VCA glue it.
- On a vocal, raise HF EXP as you add DRIVE on unit B, so the voice stays clear as it gets more compressed.
- Watch the BELOW / KNEE / ABOVE lights: for gentle levelling, the amber KNEE light should be on most of the time.
- Negative ratios sound best on sustained material and with MIX below 100%.

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